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Week 2

SESSION 3\

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.

Code:

```
UN PICO 5.09 File: 17.c Modified

printf("-----\n");
printf("Process ID (PID)      : %d\n", getpid());
printf("Parent Process ID      : %d\n", getppid());

/* 6. Project structure */
printf("\n6. Project Structure\n");
printf("-----\n");
system("ls -la");

printf("\n===== \n");
printf("          PROJECT SETUP COMPLETED\n");
printf("===== \n");

return 0;
}
```

Output:

```
nrk@nallas-mac-Book-Air ~ % gcc 17.c -o 17
nrk@nallas-mac-Book-Air ~ % ./17
=====
LINUX SHELL PROJECT SETUP
=====

1. Linux Operating System Information
-----
Darwin nallas-mac-Book-Air.local 25.5.0: Tue Jun  9 22:19:21 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8122 arm64

2. GCC Compiler Information
-----
Apple clang version 17.0.0 (clang-1700.6.3.2)
Target: arm64-apple-darwin25.5.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin

3. Git Repository Information
-----
git version 2.50.1 (Apple Git-155)

4. Shell Information
-----
Current Shell: /bin/zsh

5. Process Information
-----
Process ID (PID)      : 2234
Parent Process ID     : 2210

6. Project Structure
-----
total 3744
drwxr-xr-x  86 nrk  staff   2752 29 Aug 09:07 .
drwxr-xr-x   5 root  admin   160 18 Jul 09:32 ..
drwxr-xr-x@ 11 nrk  staff   382 28 Feb  2025 .anydesk
-rw-r--r--   1 nrk  staff   128  5 Jun 18:57 .cache
drwxr-xr-x@  4 nrk  staff    96 15 Jun 18:06 .cfuserTextEncoding
drwxr-xr-x@  3 nrk  staff    96 12 Jun 08:32 .copilot
drwxr-xr-x@  3 nrk  staff    96  1 Aug  2024 .cups
-rw-r--r--@  1 nrk  staff  18436 26 Aug 12:22 .DS_Store
drwxr-xr-x   3 nrk  staff    96 15 Jun 18:06 .idarc
drwxr-xr-x@  4 nrk  staff   128 16 Jun 14:05 .lina
drwxr-xr-x@  3 nrk  staff    96  4 Feb  2026 .local
-rw-r--r--   1 nrk  staff    62 25 Aug 11:49 .mysql_history
drwxr-xr-x@  6 nrk  staff   192  2 Jun 12:47 .npm
drwxr-xr-x@  3 nrk  staff    96 16 Jun 12:10 .ssh
drwxr-xr-x@  4 nrk  staff   128  2 Jun 12:49 .th-client
drwxr-xr-x@  4 nrk  staff   128 12 Apr 12:24 .thumbnails
drwxr-xr-x@  6 nrk  staff   192 29 Aug 08:26 .Trash
drwxr-xr-x@  5 nrk  staff   160  4 Feb  2026 .vscode
drwxr-xr-x@  3 nrk  staff    96  2 Jun 12:36 .vscode-shared
-rw-r--r--   1 nrk  staff   537 25 Aug 11:47 .zprofile
-rw-r--r--   1 nrk  staff   166 15 Jun 10:05 .zprofile.pysave
-rw-r--r--   1 nrk  staff   4059 29 Aug 09:05 .zsh_history
drwxr-xr-x@ 62 nrk  staff  1984 29 Aug 09:06 .zsh_sessions
-rw-r--r--   1 nrk  staff  1447 26 Aug 12:06 .zsh.save
drwxr-xr-x   1 nrk  staff  33472 29 Aug 08:36 11
-rw-r--r--   1 nrk  staff    802 29 Aug 08:35 11.c
-rwxr-xr-x   1 nrk  staff  33608 29 Aug 08:47 13
-rw-r--r--   1 nrk  staff   1479 29 Aug 08:47 13.C
-rwxr-xr-x   1 nrk  staff  33728 29 Aug 08:51 14
```

```
-rw-r--r-- 1 nzk staff 1479 29 Aug 08:47 13.c
-rwxr-xr-x 1 nzk staff 33728 29 Aug 08:51 14
-rw-r--r-- 1 nzk staff 2349 29 Aug 08:51 14.c
-rwxr-xr-x 1 nzk staff 33856 29 Aug 08:57 16
-rw-r--r-- 1 nzk staff 1184 29 Aug 08:57 16.c
-rwxr-xr-x 1 nzk staff 34112 29 Aug 09:01 16
-rw-r--r-- 1 nzk staff 2936 29 Aug 09:01 16.c
-rwxr-xr-x 1 nzk staff 33680 29 Aug 09:07 17
-rw-r--r-- 1 nzk staff 1665 29 Aug 09:06 17.c
-rwxr-xr-x 1 nzk staff 33584 26 Aug 12:11 2
-rw-r--r-- 1 nzk staff 1990 26 Aug 12:11 2.c
-rwxr-xr-x 1 nzk staff 33936 26 Aug 12:16 3
-rw-r--r-- 1 nzk staff 1142 26 Aug 12:14 3.c
-rwxr-xr-x 1 nzk staff 33760 26 Aug 12:20 4
-rw-r--r-- 1 nzk staff 4517 26 Aug 12:26 4.c
-rwxr-xr-x 1 nzk staff 33968 25 Aug 12:04 444
-rw-r--r-- 1 nzk staff 1282 25 Aug 12:03 444.c
-rwxr-xr-x 1 nzk staff 33760 26 Aug 12:28 5
-rw-r--r-- 1 nzk staff 1016 26 Aug 12:28 5.c
-rw----- 1 nzk staff 1394 26 Aug 11:57 a.c.save
drwxr-xr-x@ 4 nzk staff 128 16 Jul 17:12 airline-lost-found
-rw-r--r-- 1 nzk staff 1279292 24 May 19:31 ancient_dress.fbx
dcwx-----@ 4 nzk staff 128 22 Mar 2025 Applications
-rw----- 1 nzk staff 1393 26 Aug 11:58 b.c.save
drwxr-xr-x@ 7 nzk staff 224 9 Jul 15:21 blenderkit_data
-rw-r--r--@ 1 nzk staff 4636 4 Jun 08:22 class AVLTree (.java
-rw-r--r--@ 1 nzk staff 1035 4 Jun 10:41 class RankNode (.java
drwxrwxrwx 2 nzk staff 64 22 Jun 13:04 DaVinci Resolve Media
-rw-r--r--@ 1 nzk staff 7151 29 Nov 2025 ddcc.circ
dcwx-----+ 4 nzk staff 128 12 Aug 08:37 Desktop
drwx-----+ 48 nzk staff 1280 29 Aug 09:06 Documents
drwx----- 98 nzk staff 3136 29 Aug 09:03 Downloads
drwxr-xr-x@ 3 nzk staff 94 15 Jun 10:35 EXAMINATION SEATING ARRANGEMENT
-rw-r--r--@ 1 nzk staff 317 4 Feb 2026 fed.css .h
-rw-r--r-- 1 nzk staff 2 25 Aug 12:00 file.c
-rw-r--r-- 1 nzk staff 16 25 Aug 12:03 fl.c
-rw-r--r--@ 1 nzk staff 2593 4 Jun 10:21 import java.util.java
-rw-r--r-- 1 nzk staff 1450 29 Aug 08:30 kl.c
drwx-----@ 180 nzk staff 3200 28 Jul 09:09 Library
-rwxr-xr-x 1 nzk staff 33664 29 Aug 08:33 lk
-rw-r--r-- 1 nzk staff 1493 29 Aug 08:32 lk.c
-rw-r--r--@ 1 nzk staff 2215 3 Nov 2025 logism.circ
dcwx----- 8 nzk staff 286 11 Jul 14:43 Movies
dcwx-----+ 5 nzk staff 160 9 Jul 15:22 Music
-rw-r--r-- 1 nzk staff 1777 26 Aug 12:08 n.c
drwxr-xr-x@ 2 nzk staff 64 12 Apr 15:36 New Folder
-rw-r--r--@ 1 nzk staff 3929 3 Jun 12:34 package com.mediflow.java
dcwx-----+ 5 nzk staff 160 21 Jul 11:34 Pictures
-rw-r--r--@ 1 nzk staff 917 23 Feb 2026 project.html
drwxr-xr-x+ 4 nzk staff 128 31 Jul 2024 Public
-rw-r--r--@ 1 nzk staff 2829 4 Jun 10:19 public class SegmentTreeLazy (.java
-rw-r--r-- 1 nzk staff 1777 26 Aug 11:59 r.c
drwxr-xr-x@ 5 nzk staff 160 15 Jun 10:17 rohith
-rw-r--r-- 1 nzk staff 1203 26 Aug 12:05 rohith.c
-rw----- 1 nzk staff 1393 26 Aug 11:56 space.save
-rw-r--r--@ 1 nzk staff 317 4 Feb 2026 student.html.h
-rw-r--r--@ 1 nzk staff 2604 16 Feb 2026 student_registration_form.html
-rw-r--r--@ 1 nzk staff 317 4 Feb 2026 student.html.h
-rw-r--r--@ 1 nzk staff 1884 3 Jun 12:32 Untitled-1.cs
```

```
=====
PROJECT SETUP COMPLETED
=====
nzk@nallas-mac-Book-Air ~ % █
```

Session 4

Task 1: To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output

Code:

```
UW PICO 5.09                                     File: 19.c                                     Modified
```

```
printf("\n=====\\n");
printf("      TOKEN VALIDATION\\n");
printf("=====\\n");

if (validate_tokens(tokens, token_count))
{
    printf("Token stream: VALID\\n");
}
else
{
    printf("Token stream: INVALID\\n");
}

printf("\\nParsing completed successfully.\\n");

return 0;
}
```

```
AG Get Help      WC WriteOut      RR Read File      PP Prev Pg      CC Cut Text
PX Exit          PC Justify      RW Where is      PL Next Pg      CL Cur Pos
                                     To Spell
```

Output:

```
nrk@nallas-mac-Book-Air ~ % gcc 19.c -o 19
nrk@nallas-mac-Book-Air ~ % ./19
=====
SHELL TOKENIZER
=====
Enter a command:
Example: ls -l | grep txt > output.txt

Input: ls

=====
TOKEN STREAM
=====
Token 1:
  Type : WORD
  Value : "ls"

=====
TOKEN VALIDATION
=====
Token stream: VALID

Parsing completed successfully.
nrk@nallas-mac-Book-Air ~ %
```

Task 2: To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

Code:

```
UW PIDO 5.09 File: 20.c Modified

    free_tree(tree);
    return 1;
}

display_execution_structure(
    commands,
    command_count
);

printf("\n===== \n");
printf("      PARSING COMPLETED\n");
printf("===== \n");

/* Free memory */
free_commands(commands, command_count);
free_tree(tree);

return 0;
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

Output:

```
Last login: Sat Aug 29 09:10:32 on ttys000
nrk@nallas-mac-Book-Air ~ % nano 20.c

nrk@nallas-mac-Book-Air ~ % gcc 20.c -o 20
nrk@nallas-mac-Book-Air ~ % ./20
=====
SIMPLE SHELL PARSER
=====
Enter command:
Example: ls -l | grep txt > output.txt
parser> 
```



```
Last login: Sat Aug 29 08:59:57 on ttys000
nrk@nallas-mac-Book-Air ~ % nano 16.c
```

```
nrk@nallas-mac-Book-Air ~ % gcc 16.c -o 16
nrk@nallas-mac-Book-Air ~ % ./16
```

```
=====
KEYBOARD INPUT HANDLING
=====
Type a command and press ENTER.
Use BACKSPACE to delete characters.
Type 'exit' to quit.
```

```
myshell> █
```